

A Controlled Empirical Comparison of Classical and Quantum Kernel SVMs for Breast Cancer Classification

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Abstract

Quantum kernel methods map classical feature vectors into quantum state spaces, theoretically offering expressive representational capacity. However, rigorous empirical comparisons against properly tuned classical baselines under leakage-free evaluation protocols remain sparse. In this work, we conduct a controlled empirical comparison of classical Support Vector Machines (Linear and RBF kernels) and Quantum Support Vector Classifiers (QSVC) on the Wisconsin Diagnostic Breast Cancer benchmark. To operate within current noisy intermediate-scale quantum constraints, input features are projected to two and four dimensions using principal component analysis and encoded via parameterized two- and four-qubit second-order Pauli-Z feature maps. The experimental pipeline enforces strict isolation of training and testing data across five outer random splits, nested five-fold cross-validation for hyperparameter tuning, feature-map architectural ablation, and sample-size scaling. Across all evaluated dimensions, classical SVMs consistently outperform QSVC in malignant-class F1 score (0.934 vs. 0.868 in two dimensions; 0.949 vs. 0.872 in four dimensions) and classification accuracy (0.951 vs. 0.907; 0.963 vs. 0.905), winning on 5/5 outer test splits. Feature-map ablation reveals that QSVC performance degrades sharply with circuit depth, while sample-size scaling demonstrates no small-data quantum advantage. Furthermore, centered kernel alignment confirms that although the four-qubit quantum kernel induces a geometry substantially distinct from classical RBF similarity ($\text{CKA} \approx 0.338$), this novelty does not translate into superior classification boundaries. While exact statevector simulation incurs substantial computational overhead, five overlapping outer splits limit formal asymptotic statistical inference. Under the evaluated conditions, no quantum advantage is observed.

Keywords: Quantum machine learning, Support Vector Classifier, Quantum kernel methods, ZZ feature map, Wisconsin Diagnostic Breast Cancer, Controlled empirical benchmark.

1 Introduction

Kernel methods in machine learning operate by implicitly mapping input vectors from a classical data space into a higher-dimensional reproducing kernel Hilbert space (RKHS), where non-linear decision boundaries can be identified via convex quadratic programming [1, 2]. In recent years, quantum machine learning (QML) has extended this paradigm by encoding classical data vectors into quantum states via parameterized quantum circuits [3, 4]. By evaluating transition fidelities between prepared quantum states, a quantum processor evaluates a quantum kernel

Gram matrix that may be classically intractable to compute, providing a theoretical foundation for potential quantum advantage in classification tasks [5–7].

Despite theoretical promise, translating quantum kernel formulations into measurable empirical advantages on real-world datasets presents profound methodological challenges [8–10]. Many published studies evaluate quantum classifiers on standard tabular medical diagnostic datasets—most notably the Wisconsin Diagnostic Breast Cancer (WDBC) benchmark—under simplified or flawed conditions [10–12]. Frequently, these demonstrations rely on a single train/test split, omit classical hyperparameter tuning, report default classical baselines, or inadvertently introduce data leakage by fitting dimensionality reduction techniques on the entire dataset prior to partitioning [8, 13]. Furthermore, the sensitivity of quantum kernels to circuit depth and entanglement topology is often overlooked, and the substantial computational overhead of evaluating quantum kernels relative to classical optimization is rarely quantified in wall-clock terms.

To address these methodological shortcomings, this study presents an end-to-end controlled empirical benchmark comparing classical SVMs (Linear and Radial Basis Function kernels) against Quantum Support Vector Classifiers (QSVC with parameterized ZZ feature maps). We examine classification performance across two- and four-dimensional representations of the WDBC dataset under a strict nested cross-validation protocol evaluated across five fixed outer random splits. We systematically investigate feature-map depth and entanglement ablation, sample-size scaling dynamics, kernel geometric alignment, and computational execution costs.

Our central empirical finding is unequivocal: under the evaluated conditions, no quantum advantage was observed for the tested dataset, preprocessing pipeline, qubit scale, and ZZ feature-map family. Rather than framing this outcome as a universal indictment of quantum machine learning, we report this as a valid, reproducible negative result that clarifies the practical boundaries of quantum kernel methods on low-dimensional tabular data.

1.1 Summary of Contributions

To provide an authentic, hype-free methodological baseline for quantum machine learning on tabular data, this work delivers six concrete contributions:

1. **Leakage-Free Nested Protocol:** Implementation of strict outer test quarantine with fold-specific refitting of all preprocessing transformers (`StandardScaler`, `PCA`, `MinMaxScaler`) and quantum Gram matrices across five outer random splits, eliminating subtle data leakage.
2. **Multi-Seed Paired Robustness:** Paired split-level evaluation across five matching random seeds ([42, 123, 456, 789, 2026]) with exact Wilcoxon signed-rank tests, Holm-Bonferroni multiplicity corrections, and exploratory split-level percentile bootstrap intervals.
3. **Controlled Feature-Map Ablation (60 Quantum Runs):** Systematic isolation of circuit depth ($\text{reps} \in \{1, 2, 3\}$) and entanglement topology (`linear` vs. `full`) across both 2-qubit and 4-qubit scales, identifying circuit depth as the primary driver of performance collapse in 4 qubits.
4. **Empirical Sample-Size Scaling Analysis:** Direct empirical testing of the “small-data quantum advantage” conjecture across training subsets ($N_{\text{train}} \in \{50, 100, 200, 300, 455\}$), showing that QSVC experiences its greatest performance deficit in low-sample regimes.
5. **Operator Feature Space and CKA Geometric Analysis:** Theoretical clarification of the 4^n -dimensional density-operator feature space and geometric characterization via Centered Kernel Alignment (CKA) and spectral entropy, proving that geometric departure from classical RBF kernels does not yield superior classification boundaries.

6. **Transparent, Reproducible Negative Benchmark:** A permanently preserved, frozen v1.0.0 benchmark suite with open-source code, execution manifests, automated test suites, and raw per-fold logs establishing verifiable empirical bounds for NISQ-era classifiers.

2 Research Questions

To guide this controlled comparative investigation, we establish eight explicit research questions:

- **RQ1 (Accuracy):** Does the evaluated quantum kernel improve classification accuracy relative to tuned classical SVM baselines?
- **RQ2 (Diagnostic Metrics):** How do malignant-class F1 score and ROC-AUC compare between classical and quantum kernel machines?
- **RQ3 (Dimensionality Scaling):** How does QSVC behavior change when expanding from two to four PCA dimensions and corresponding qubits?
- **RQ4 (Computational Scaling):** How does computational cost change as the qubit count and feature dimension double?
- **RQ5 (Quantum Kernel Geometry):** What similarity geometry, off-diagonal distribution, and spectral effective rank are induced by the quantum feature map?
- **RQ6 (Geometric Alignment):** How substantially does the quantum kernel geometry depart from the classical RBF similarity geometry as measured by Centered Kernel Alignment (CKA)?
- **RQ7 (Computational Overhead):** What is the exact computational overhead of statevector quantum kernel simulation relative to classical CPU SVM optimization?
- **RQ8 (Split Stability):** Are the observed empirical rankings stable across multiple independent random train/test partitions?

3 Related Work

The literature contextualizing quantum kernel classifiers spans classical statistical learning theory, quantum circuit expressivity, kernel geometry, and empirical benchmarking rigor.

3.1 Classical Kernel Methods and Inductive Biases

Classical Support Vector Machines (SVMs) formulate pattern classification as the identification of a maximal-margin separating hyperplane in an inner-product space [1, 2]. When patterns are not linearly separable in their original representation, Mercer kernels $k(\mathbf{x}, \mathbf{x}') = \langle \phi(\mathbf{x}), \phi(\mathbf{x}') \rangle_{\mathcal{H}}$ implicitly project inputs into a reproducing kernel Hilbert space $\mathcal{H}$ without requiring explicit coordinate evaluation [14]. For tabular and continuous biological datasets, Linear and Gaussian Radial Basis Function (RBF) kernels provide well-characterized inductive biases [13]. Classical SVM optimization is solved via dual convex quadratic programming, where libraries such as LibSVM implement highly efficient Sequential Minimal Optimization (SMO) routines scaling between $\mathcal{O}(N^2)$ and $\mathcal{O}(N^3)$ in sample size [15, 16].

3.2 Quantum Feature Maps and Quantum Kernel Formulations

Quantum kernel methods replace classical non-linear feature maps with quantum state preparation unitaries $U_{\Phi}(\mathbf{x})$ acting on an n -qubit reference state $|0^{\otimes n}\rangle$ [3, 4]. This operation maps a classical vector $\mathbf{x} \in \mathbb{R}^d$ to a quantum pure state $|\psi(\mathbf{x})\rangle$ or density operator $\rho(\mathbf{x}) = |\psi(\mathbf{x})\rangle\langle\psi(\mathbf{x})|$. The corresponding kernel is evaluated as the quantum transition fidelity:

$$K(\mathbf{x}, \mathbf{x}') = |\langle\psi(\mathbf{x})|\psi(\mathbf{x}')\rangle|^2 = \text{Tr} [\rho(\mathbf{x})\rho(\mathbf{x}')] \quad (1)$$

Schuld demonstrated that supervised quantum classifiers trained with variational parameters are fundamentally linear models in quantum feature spaces, establishing formal equivalence between variational quantum classifiers (VQCs) and quantum kernel machines [6]. The Second-Order Pauli-Z Expansion (ZZFeatureMap) proposed by Havlíček et al. [3] encodes features via single-qubit phase gates and entangles qubits using pairwise controlled-phase rotations. This architecture derives its theoretical motivation from instantaneous quantum polynomial (IQP) circuit families, for which classical sampling of the output distribution is conjectured to be intractable under standard complexity assumptions [17, 18]. However, Hubregtsen et al. [19] observed that higher circuit entanglement does not correlate monotonically with improved classification accuracy, and alternative architectures such as data re-uploading [20] have been proposed to enhance expressivity.

3.3 Quantum Kernel Generalization, Concentration, and Untrainability

A critical development in quantum machine learning theory is the discovery of **kernel concentration**, an analog to barren plateaus in variational quantum circuits [21, 22]. Thanasi, Wang, Cerezo, and Holmes [23] proved that for deep or highly expressive parameterized quantum circuits, the fidelity kernel concentrates exponentially around a constant value as circuit depth or qubit count increases:

$$\text{Var}_{\mathbf{x}, \mathbf{x}'} [K(\mathbf{x}, \mathbf{x}')] \in \mathcal{O}\left(\frac{1}{2^n}\right) \quad (2)$$

This concentration causes the empirical Gram matrix to approach the identity matrix $\mathbf{I}$, extinguishing the classifier’s discriminative capacity and rendering dual optimization ineffective [24]. Kübler, Buchholz, and Schölkopf [25] showed that without problem-specific data alignment, quantum kernels exhibit an inductive bias that is frequently unfavorable for classical classification tasks. Furthermore, Shaydulin and Wild [26] established that feature scaling acts as an effective bandwidth hyperparameter, where improper coordinate bounds lead to artificial orthogonality or trivial state collapse.

3.4 Sample Complexity and Small-Data Regimes

A prominent hypothesis in quantum machine learning posits that quantum kernels might exhibit superior sample efficiency in small-data regimes ($N \ll 100$) where classical models risk overfitting [27]. Caro et al. [27] and Banchi et al. [28] proved rigorous generalization bounds within the quantum Probably Approximately Correct (PAC) framework, showing that quantum models can generalize from limited samples when generator norms or effective circuit dimensions are constrained. Canatar et al. [29] analyzed this phenomenon through the spectral decomposition of quantum Gram matrices, proving that eigenvalue decay governs effective model capacity. However, these theoretical upper bounds describe what is learnable in principle; they do not guarantee that quantum kernels will outperform classical regularized baselines on real-world tabular data.

3.5 Biomedical Benchmarking on Oncological Data

The Wisconsin Diagnostic Breast Cancer (WDBC) dataset [30] has frequently served as an experimental testbed for evaluating emerging quantum classifiers [11, 12]. Wang [12] proposed a multi-objective genetic algorithm for feature selection with QSVMs, reporting high accuracies on selected feature subsets. Azevedo, Silva, and Dutra [11] evaluated quantum transfer learning on breast cancer images, finding performance on par with classical linear models. More recently, Leither, Lubinski, Kubal, and Johri [10] conducted a comprehensive benchmark across diverse oncological datasets using the Red Cedar framework against AutoML classical neural networks, finding **no empirical quantum advantage** across evaluated cancer classification tasks. Notably, many prior studies on WDBC evaluated a single train/test split without nested cross-validation or leakage isolation, making direct performance comparisons methodologically fraught.

3.6 Empirical Rigor and Quantum Advantage Claims

Establishing genuine quantum advantage in machine learning requires navigating stringent theoretical and empirical hurdles [9, 31]. Aaronson [32] emphasized that quantum speedup claims frequently rely on input/output caveats, where classical data loading overheads negate computational gains. Tang [33] demonstrated that classical “dequantization” algorithms can match quantum speedups when given comparable randomized access. In a comprehensive critique across hundreds of datasets, Bowles, Ahmed, and Schuld [8] revealed that apparent quantum advantages in the literature frequently evaporate when classical baselines are properly tuned and data leakage is eliminated. Rigorous empirical benchmarking demands that quantum algorithms be tested under identical partitioning, leakage-free preprocessing, and multi-seed robustness protocols.

4 Dataset and Problem Formulation

The empirical evaluation is conducted on the **Wisconsin Diagnostic Breast Cancer (WDBC)** dataset, originally compiled by Street, Wolberg, and Mangasarian [30] and distributed through scikit-learn [16].

- **Sample Size:** $N = 569$ patient observations.
- **Feature Space:** 30 continuous real-valued features computed from digitized images of fine needle aspirates (FNA) of breast masses, describing characteristics of cell nuclei (e.g., radius, texture, perimeter, area, smoothness, compactness, concavity, concave points, symmetry, and fractal dimension across mean, standard error, and “worst” measurements).
- **Target Diagnostic Classes:** Binary classification:
 - **Malignant:** 212 samples (37.26%).
 - **Benign:** 357 samples (62.74%).
- **Positive Class Designation:** In accordance with oncological screening priority, **Malignant (Class 0)** is treated as the primary positive class (`pos_label=0`) across all precision, recall, F1, and ROC-AUC evaluations. Decision threshold rankings and probability scores are oriented strictly toward detecting malignancy.
- **Methodological Scope Disclaimer:** This investigation is strictly designed as a methodological machine-learning benchmark evaluating kernel representations. It does **not** constitute a clinical validation study, and the resulting models are not intended for medical diagnostic decision-making.

5 Experimental Protocol and Leakage Prevention

To ensure absolute methodological integrity and eliminate data leakage, the experimental architecture employs an isolated outer split design with nested inner cross-validation across five fixed random seeds: `SEEDS = [42, 123, 456, 789, 2026]` [8].

5.1 Partitioning Protocol

1. **Outer Split:** An 80/20 stratified partition divides the 569 samples into 455 training observations and 114 test observations. Outer test partitions remain completely quarantined during all preprocessing fitting and hyperparameter selection.
2. **Inner Cross-Validation:** Within each outer training set ($N = 455$), a 5-fold stratified cross-validation is performed (364 training, 91 validation per fold).
3. **Leakage Protection:** `StandardScaler`, `PCA`, and quantum `MinMaxScaler` are fitted exclusively on the training split of the respective partition [16]. They are applied to validation and test partitions strictly via `transform()`. In the inner tuning phase, transformers and quantum Gram matrices are independently recomputed inside each fold.

5.2 Dimensionality Reduction

Due to qubit scalability constraints in current quantum simulators and NISQ devices [31], the 30 standardized features are compressed using Principal Component Analysis (PCA):

- **PCA 2:** 2 principal components capturing $63.5\% \pm 0.6\%$ of cumulative feature variance.
- **PCA 4:** 4 principal components capturing $79.4\% \pm 0.4\%$ of cumulative feature variance.

6 Classical Support Vector Machine Baselines

Classical benchmarks utilize the standard LibSVM implementation via scikit-learn [15, 16]:

1. **Linear SVM (`SVC(kernel='linear')`):** Linear decision hyperplane optimized over regularization parameter $C \in \{0.01, 0.1, 1.0, 10.0, 100.0\}$ [1].
2. **Radial Basis Function (RBF) SVM (`SVC(kernel='rbf')`):** Non-linear Gaussian kernel $K(\mathbf{x}, \mathbf{x}') = \exp(-\gamma \|\mathbf{x} - \mathbf{x}'\|^2)$ optimized over $C \in \{0.01, 0.1, 1.0, 10.0, 100.0\}$ and $\gamma \in \{\text{'scale'}, \text{'auto'}, 0.01, 0.1, 1.0\}$ [2, 13].
3. **Inner-Selected Classical Comparator:** To prevent selective reporting bias, the classical model family (Linear vs. RBF) and hyperparameter configuration achieving the highest mean inner-CV malignant F1 score is designated as the canonical classical comparator for that outer split [8]. Ties are resolved deterministically by favoring smaller C , followed by Linear over RBF.

7 Quantum Kernel Support Vector Classifier (QSVC)

7.1 Quantum Feature Map Formulation

The quantum classifier maps classical feature vectors $\mathbf{x} \in [0, \pi]^n$ into an n -qubit quantum state space using the Second-Order Pauli-Z Expansion (`zz_feature_map`) implemented in Qiskit [18, 34]:

$$\mathcal{U}_{\Phi}(\mathbf{x}) = \prod_{d=1}^{\text{reps}} \left(\prod_{i < j} U_{\Phi_{i,j}}(\mathbf{x}) \prod_{k=1}^n U_{\Phi_k}(\mathbf{x}) H^{\otimes n} \right) \quad (3)$$

where $H^{\otimes n}$ denotes a layer of Hadamard gates initializing an equal superposition, $U_{\Phi_k}(\mathbf{x}) = \exp(-i\Phi_k(\mathbf{x})Z_k/2)$ encodes individual features via single-qubit phase rotations with $\Phi_k(\mathbf{x}) = 2x_k$, and $U_{\Phi_{i,j}}(\mathbf{x}) = \exp(-i\Phi_{i,j}(\mathbf{x})Z_iZ_j/2)$ encodes pairwise non-linear interactions via two-qubit controlled-phase gates with:

$$\Phi_{i,j}(\mathbf{x}) = 2(\pi - x_i)(\pi - x_j) \quad (4)$$

As demonstrated by Shaydulin and Wild [26], mapping input components to $[0, \pi]$ sets an effective bandwidth scale for the statevector encoding.

7.2 Fidelity Kernel and Operator Feature Space

The similarity between two classical samples $\mathbf{x}_i$ and $\mathbf{x}_j$ is defined by the quantum state transition fidelity [3, 4]:

$$K(\mathbf{x}_i, \mathbf{x}_j) = |\langle \psi(\mathbf{x}_i) | \psi(\mathbf{x}_j) \rangle|^2 = \text{Tr} [\rho(\mathbf{x}_i) \rho(\mathbf{x}_j)] \quad (5)$$

where $\rho(\mathbf{x}) = |\psi(\mathbf{x})\rangle\langle\psi(\mathbf{x})|$ is the pure state density operator.

Operator Feature Space Formulation: As demonstrated by Schuld [6], although the underlying quantum statevectors $|\psi(\mathbf{x})\rangle$ reside in an n -qubit complex Hilbert state space $\mathcal{H} = \mathbb{C}^{2^n}$ of dimension 2^n , the fidelity kernel mathematically evaluates the Hilbert-Schmidt inner product of density operators in the linear operator space $\mathcal{B}(\mathcal{H}) \cong \mathcal{H} \otimes \mathcal{H}^*$. The linear dimension of this operator feature space scales as:

$$\dim \mathcal{B}(\mathcal{H}) = (2^n)^2 = 4^n \quad (6)$$

(or $4^n - 1$ for the subspace of traceless Hermitian operators). Consequently, the maximum algebraic rank of the fidelity Gram matrix is $\min(N, 4^n)$ —scaling up to 16 for $n = 2$ qubits ($4^2 = 16$) and up to 256 for $n = 4$ qubits ($4^4 = 256$)—rather than being bounded by the statevector dimension 2^n .

7.3 Statevector Simulation Engine

To establish an exact, noise-free theoretical upper bound on classification performance, kernel matrices are evaluated via exact statevector linear algebra on classical CPUs:

$$\mathbf{K} = |\Psi \Psi^\dagger|^2 \quad (7)$$

where $\Psi \in \mathbb{C}^{N \times 2^n}$ represents the matrix of statevectors. Gram matrix diagonals are explicitly enforced to $K_{i,i} = 1.0$, and numerical symmetry and positive semi-definiteness (eigvals $\geq -10^{-10}$) are verified.

Simulation vs. Hardware Distinction: Exact statevector simulation evaluates mathematical inner products directly without shot noise, gate infidelities, or decoherence [35]. It provides an ideal representation of the feature-map geometry but does **not** represent physical quantum processor (QPU) execution latencies.

7.4 Canonical QSVC Configuration

Based on controlled architectural ablation (Section 8), the canonical QSVC configuration is frozen at:

- **Two-Qubit (2Q):** PCA components = 2, qubits = 2, reps = 1, entanglement = 'full'.
- **Four-Qubit (4Q):** PCA components = 4, qubits = 4, reps = 1, entanglement = 'full'.
- Regularization parameter $C \in \{0.01, 0.1, 1.0, 10.0, 100.0\}$ is tuned via inner cross-validation.

8 Quantum Feature-Map Ablation Study

To understand the sensitivity of quantum kernel representations to circuit depth and entanglement topology, we conducted an exhaustive 60-run ablation experiment across circuit repetitions ($\text{reps} \in \{1, 2, 3\}$), entanglement topologies ('linear' vs. 'full'), and qubit dimensions ($q \in \{2, 4\}$) across all five outer splits [19].

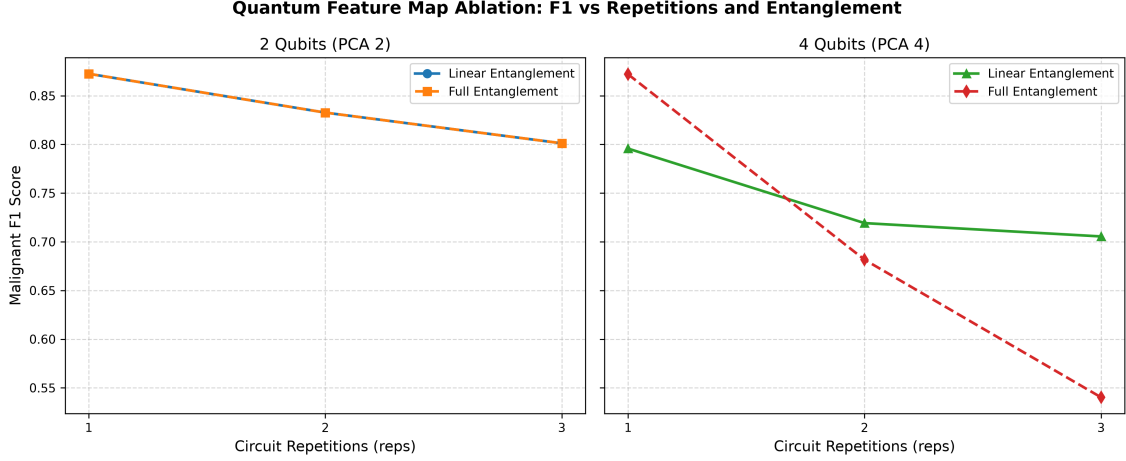


Figure 1: **Figure 2: Quantum Feature-Map Ablation Dynamics.** Impact of circuit repetitions ($\text{reps} \in \{1, 2, 3\}$) and entanglement topology (linear vs. full) on outer-test malignant F1 score across 2-qubit and 4-qubit representations. Circuit depth is the primary driver of performance degradation in 4 qubits.

Table 1: Quantum Feature-Map Ablation across 5 Outer Splits ($n = 60$ Quantum Runs, Mean $\pm$ SD).

Qubits (q)	Reps	Entanglement	Outer Malignant F1	Outer ROC-AUC	Outer Accuracy	Off-Diag Mean	Off-Diag Std	Effective Rank
2	1	Linear	0.8726 ± 0.0376	0.9697 ± 0.0141	0.9123 ± 0.0224	0.3244 ± 0.0071	0.2908 ± 0.0039	7.35 ± 0.21
2	1	Full	0.8726 ± 0.0376	0.9697 ± 0.0141	0.9123 ± 0.0224	0.3244 ± 0.0071	0.2908 ± 0.0039	7.35 ± 0.21
2	2	Linear	0.8327 ± 0.0283	0.9414 ± 0.0240	0.8842 ± 0.0169	0.4311 ± 0.0302	0.2795 ± 0.0033	6.33 ± 0.50
2	2	Full	0.8327 ± 0.0283	0.9414 ± 0.0240	0.8842 ± 0.0169	0.4311 ± 0.0302	0.2795 ± 0.0033	6.33 ± 0.50
2	3	Linear	0.8012 ± 0.0352	0.9204 ± 0.0250	0.8596 ± 0.0215	0.4818 ± 0.0249	0.2677 ± 0.0015	5.90 ± 0.40
2	3	Full	0.8012 ± 0.0352	0.9204 ± 0.0250	0.8596 ± 0.0215	0.4818 ± 0.0249	0.2677 ± 0.0015	5.90 ± 0.40
4	1	Linear	0.7957 ± 0.0688	0.9263 ± 0.0303	0.8596 ± 0.0345	0.0818 ± 0.0041	0.1306 ± 0.0072	70.29 ± 4.94
4	1	Full (Canonical)	0.8721 ± 0.0556	0.9581 ± 0.0227	0.9053 ± 0.0423	0.0961 ± 0.0032	0.1015 ± 0.0046	96.49 ± 5.41
4	2	Linear	0.7192 ± 0.0859	0.8935 ± 0.0442	0.8140 ± 0.0423	0.0829 ± 0.0035	0.1057 ± 0.0046	95.80 ± 5.39
4	2	Full	0.6816 ± 0.0590	0.8546 ± 0.0482	0.7789 ± 0.0364	0.0811 ± 0.0015	0.0845 ± 0.0019	123.66 ± 2.65
4	3	Linear	0.7055 ± 0.0493	0.8700 ± 0.0311	0.7965 ± 0.0413	0.0977 ± 0.0025	0.1080 ± 0.0031	93.63 ± 4.01
4	3	Full	0.5403 ± 0.0523	0.7872 ± 0.0340	0.7228 ± 0.0295	0.0778 ± 0.0018	0.0770 ± 0.0017	137.43 ± 2.83

8.1 Key Empirical Findings

1. **Two-Qubit Regimes:** In 2Q, linear and full entanglement are algebraically identical because only a single qubit pair (0, 1) exists. As circuit depth increases, outer-test malignant F1 degrades monotonically (reps = 1: 0.8726; reps = 2: 0.8327; reps = 3: 0.8012).
2. **Four-Qubit Regimes:** In 4Q, circuit depth exhibits an acute interaction with entanglement. Full entanglement with reps = 1 achieves the strongest performance (0.8721 ± 0.0556), whereas deeper configurations collapse severely (reps = 3, Full: 0.5403 ± 0.0523).

8.2 Conservative Methodological Interpretation

The severe degradation observed in deeper 4Q configurations ($\text{reps} \geq 2$) in full entanglement was strongly associated with circuit depth. Reducing depth to $\text{reps} = 1$ substantially restored 4Q performance from 0.6816 to 0.8721. We refrain from asserting that circuit depth alone caused model collapse; rather, deeper feature-map configurations across the 256-dimensional operator feature space were strongly associated with degraded predictive performance and excessive dimensional dispersion, displaying empirical characteristics consistent with phenomena analyzed in quantum kernel concentration literature [22, 23].

9 Sample-Size Scaling Dynamics

To test the hypothesis that quantum kernels provide an inductive bias advantage in small-data regimes [27, 28], we evaluated model performance across nested training subsets $N_{\text{train}} \in \{50, 100, 200, 300, 455\}$ with preprocessing strictly refit on each subset and evaluated on fixed held-out test sets ($N_{\text{test}} = 114$).

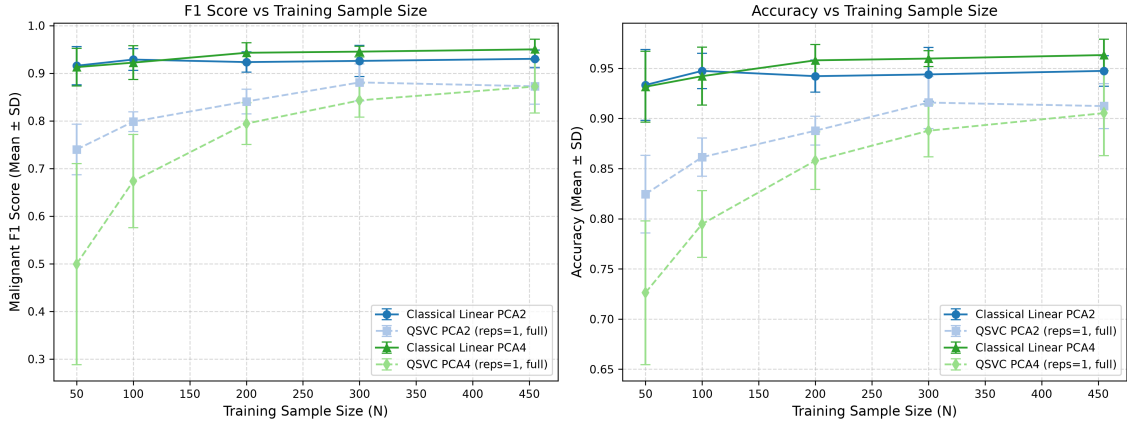


Figure 2: **Figure 3: Sample-Size Scaling Dynamics.** Mean malignant F1 score across training subset sizes $N_{\text{train}} \in \{50, 100, 200, 300, 455\}$. Classical SVMs maintain a consistent performance advantage across all sample regimes, with QSVc exhibiting its largest deficit at $N = 50$.

9.1 Empirical Observations

- **Absence of Small-Data Quantum Advantage:** At $N_{\text{train}} = 50$, classical models retained robust diagnostic discrimination:
 - Classical Linear SVM (PCA 2): $\text{F1} = 0.9157 \pm 0.0402$, $\text{Accuracy} = 0.9333 \pm 0.0354$
 - Classical Linear SVM (PCA 4): $\text{F1} = 0.9128 \pm 0.0395$, $\text{Accuracy} = 0.9316 \pm 0.0353$
 - QSVc (2Q): $\text{F1} = 0.7398 \pm 0.0531$ (performance deficit: -0.1759)
 - QSVc (4Q): $\text{F1} = 0.4994 \pm 0.2110$ (performance deficit: -0.4134 , with recall collapsing to 0.4095)
- **Sample Efficiency Convergence:** As training size expanded from $N = 50$ to $N = 455$, QSVc performance steadily improved (2Q: $0.740 \rightarrow 0.873$; 4Q: $0.499 \rightarrow 0.872$). However, classical SVMs maintained a substantial lead at every evaluated sample size. Full tabular scaling values are documented in Supplementary Table S1.

10 Canonical Predictive Results

Final canonical performance metrics across the five frozen outer test splits are summarized in Table 2 and visualized in Figure 3.

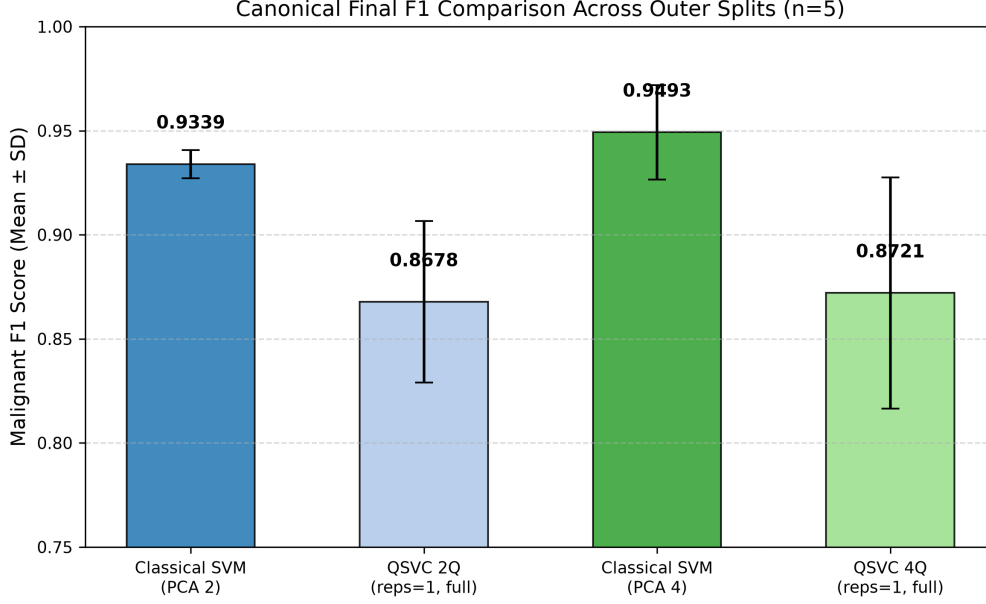


Figure 3: **Figure 1: Final Malignant F1 Score Comparison.** Mean and individual outer-split test F1 scores for Classical Comparator vs. Canonical QSVc across 2-qubit and 4-qubit representations. Classical models outperform QSVc on 5/5 splits in both representations.

Table 2: Canonical Outer-Test Performance Summary across 5 Outer Splits (Mean $\pm$ SD).

Model Architecture	Repr.	Qubits	Test Accuracy	Test Precision	Test Recall	Malignant F1	Test ROC-AUC
Classical Comparator	PCA 2	0	0.9509 $\pm$ 0.0048	0.9257 $\pm$ 0.0172	0.9429 $\pm$ 0.0213	0.9339 $\pm$ 0.0068	0.9866 $\pm$ 0.0074
Tuned Linear SVM	PCA 2	0	0.9509 $\pm$ 0.0078	0.9266 $\pm$ 0.0318	0.9429 $\pm$ 0.0213	0.9341 $\pm$ 0.0093	0.9894 $\pm$ 0.0030
Tuned RBF SVM	PCA 2	0	0.9456 $\pm$ 0.0096	0.9245 $\pm$ 0.0182	0.9286 $\pm$ 0.0238	0.9263 $\pm$ 0.0133	0.9810 $\pm$ 0.0119
Canonical QSVc	PCA 2	2	0.9070 $\pm$ 0.0237	0.9064 $\pm$ 0.0457	0.8381 $\pm$ 0.0832	0.8678 $\pm$ 0.0388	0.9644 $\pm$ 0.0244
Classical Comparator	PCA 4	0	0.9632 $\pm$ 0.0157	0.9570 $\pm$ 0.0184	0.9429 $\pm$ 0.0464	0.9493 $\pm$ 0.0227	0.9941 $\pm$ 0.0039
Tuned Linear SVM	PCA 4	0	0.9684 $\pm$ 0.0100	0.9671 $\pm$ 0.0255	0.9476 $\pm$ 0.0391	0.9565 $\pm$ 0.0143	0.9952 $\pm$ 0.0029
Tuned RBF SVM	PCA 4	0	0.9561 $\pm$ 0.0139	0.9434 $\pm$ 0.0244	0.9381 $\pm$ 0.0398	0.9401 $\pm$ 0.0198	0.9935 $\pm$ 0.0045
Canonical QSVc	PCA 4	4	0.9053 $\pm$ 0.0423	0.8748 $\pm$ 0.0767	0.8762 $\pm$ 0.0832	0.8721 $\pm$ 0.0556	0.9581 $\pm$ 0.0227

10.1 Performance Synthesis

- Primary Endpoint (Malignant F1):** In the 2-dimensional representation, the Classical Comparator achieved a Malignant F1 of 0.9339 ± 0.0068 , outperforming QSVc 2Q (0.8678 ± 0.0388). In the 4-dimensional representation, the Classical Comparator achieved an F1 of 0.9493 ± 0.0227 (with Linear SVM reaching 0.9565 ± 0.0143), while QSVc 4Q attained 0.8721 ± 0.0556 .
- Secondary Descriptive Endpoints:** Across all secondary metrics (Test Accuracy, Precision, Recall, and ROC-AUC), classical models demonstrated higher descriptive performance. In PCA 2, classical accuracy reached 0.9509 ± 0.0048 vs. 0.9070 ± 0.0237 for QSVc 2Q, with classical recall showing higher sensitivity for malignancy (0.9429 vs. 0.8381). In PCA 4, classical accuracy reached 0.9632 ± 0.0157 vs. 0.9053 ± 0.0423 for QSVc 4Q. Classical ROC-AUC exceeded 0.986 (PCA 2) and 0.994 (PCA 4), whereas QSVc reached

0.9644 (2Q) and 0.9581 (4Q). In accordance with our pre-registered statistical protocol, these secondary metrics are reported as descriptive comparative indicators and were not subjected to formal hypothesis testing.

11 Kernel Geometry Analysis

To analyze the mathematical properties of the feature spaces, we examined the training Gram matrices (455×455) using spectral decomposition [25, 29]:

$$\text{Effective Rank} = \exp \left(- \sum_{i=1}^N p_i \ln p_i \right), \quad p_i = \frac{\lambda_i}{\sum_j \lambda_j} \quad (8)$$

where λ_i denote the non-negative eigenvalues of the normalized Gram matrix.

11.1 Geometry Properties

- **Two-Qubit Quantum Kernel:** Mean off-diagonal similarity: 0.3244 ± 0.0071 ; off-diagonal std: 0.2908 ± 0.0039 ; effective rank: 7.35 ± 0.21 .
- **Four-Qubit Quantum Kernel:** Mean off-diagonal similarity: 0.0961 ± 0.0032 ; off-diagonal std: 0.1015 ± 0.0046 ; effective rank: 96.49 ± 5.41 .

11.2 Mathematical Interpretation and Operator Dimensionality

A vital theoretical distinction must be maintained between the statevector Hilbert dimension 2^n (4 for 2Q, 16 for 4Q) and the linear operator feature space $\mathcal{B}(\mathcal{H})$ whose dimension scales up to 4^n (16 for 2Q, 256 for 4Q) [4, 6]. Because the fidelity kernel evaluates the Hilbert-Schmidt inner product of density operators $K(\mathbf{x}_i, \mathbf{x}_j) = \text{Tr}[\rho(\mathbf{x}_i)\rho(\mathbf{x}_j)]$, the linear capacity of the kernel Gram matrix is bounded by 4^n rather than 2^n .

In the 2-qubit space ($4^2 = 16$ operator dimensions), statevectors maintain moderate pairwise overlap (mean similarity 0.3244 ± 0.0071) with an effective rank of 7.35 ± 0.21 , indicating that data representations occupy roughly half of the available operator capacity. In the 4-qubit space ($4^4 = 256$ operator dimensions), state representations become dispersed across a vastly expanded operator manifold: average off-diagonal fidelity drops to 0.0961 ± 0.0032 , while effective rank expands to 96.49 ± 5.41 . This substantial dispersion across the 256-dimensional operator feature space exhibits characteristics consistent with phenomena analyzed in quantum kernel concentration literature [22, 23]. We treat effective rank strictly as a descriptive diagnostic of spectral Shannon entropy in the density-operator feature space, avoiding causal claims regarding classifier untrainability.

12 Classical RBF vs. Quantum Kernel Alignment

To evaluate whether the quantum feature map creates a representation fundamentally different from classical kernels, we computed the **Centered Kernel Alignment (CKA)** between the quantum Gram matrix $\mathbf{K}$ and the seed-specific tuned classical RBF reference matrix $\mathbf{L}$ [1, 14]:

$$\text{CKA}(\mathbf{K}, \mathbf{L}) = \frac{\langle \mathbf{K}_c, \mathbf{L}_c \rangle_F}{\|\mathbf{K}_c\|_F \|\mathbf{L}_c\|_F}, \quad \mathbf{K}_c = \mathbf{H}\mathbf{K}\mathbf{H} \quad (9)$$

where $\mathbf{H} = \mathbf{I} - \frac{1}{N}\mathbf{1}\mathbf{1}^T$ is the empirical centering matrix.

Classical RBF vs Quantum Fidelity Kernel Heatmaps (Seed 42, First 50 Samples)

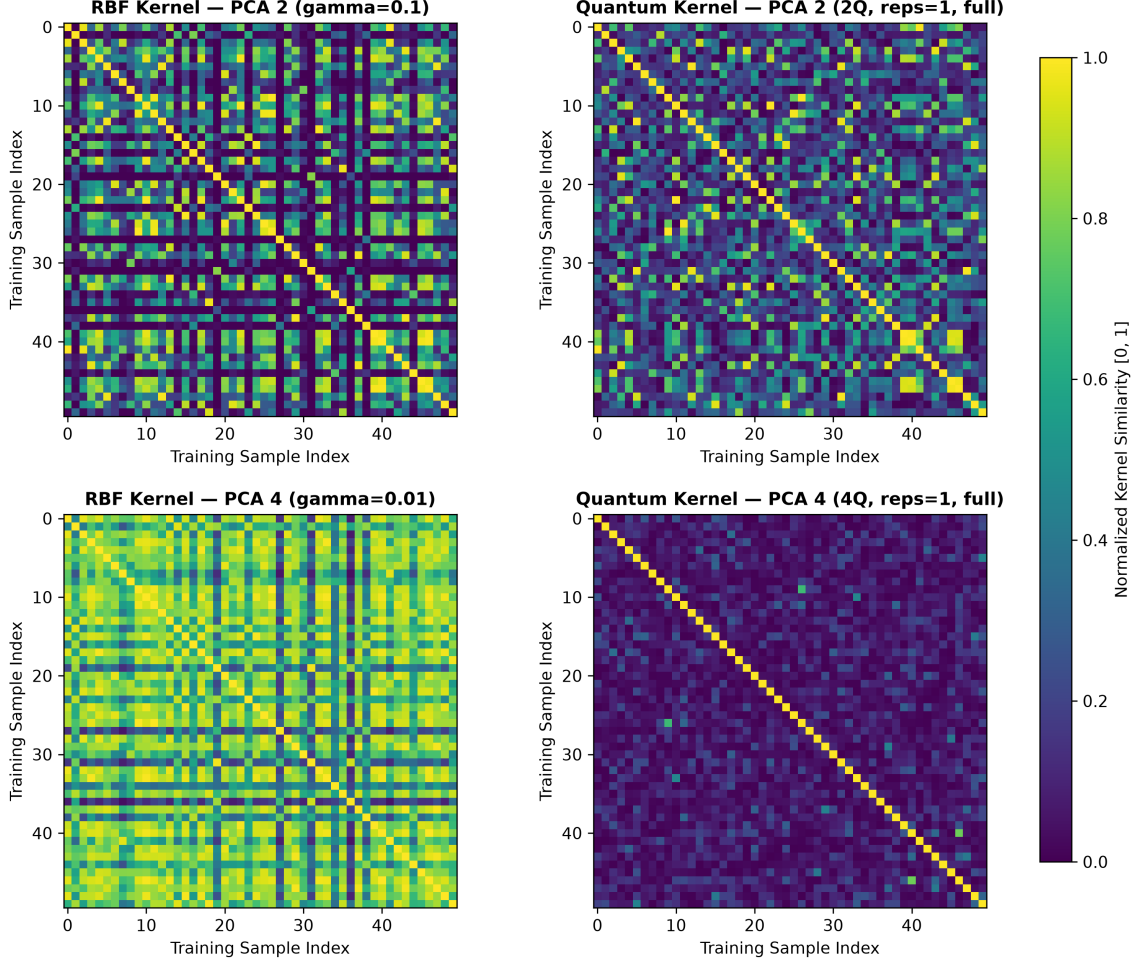


Figure 4: Classical RBF vs. Quantum Kernel Heatmaps. Training Gram matrix similarity structures (455×455) across Classical RBF, 2Q QSVC, and 4Q QSVC. The 4Q quantum kernel exhibits marked geometric departure from classical RBF similarity (CKA ≈ 0.338).

Table 3: Kernel Geometry and Centered Kernel Alignment (CKA) across 5 Outer Splits (Mean $\pm$ SD).

Repr. / Qubits	CKA Alignment	Frobenius Align.	Class. RBF Eff. Rank	Quantum Eff. Rank	Class. RBF Off-Diag	Quantum Off-Diag
PCA 2 / 2Q	0.5732 ± 0.1578	0.8344 ± 0.0327	6.52 ± 3.86	7.35 ± 0.21	0.4895 ± 0.2396	0.3244 ± 0.0071
PCA 4 / 4Q	0.3375 ± 0.0718	0.7162 ± 0.0119	6.97 ± 5.06	96.49 ± 5.41	0.4765 ± 0.1901	0.0961 ± 0.0032

12.1 Alignment Results and Scientific Rationale

- **PCA 2 / 2Q:** Mean CKA = 0.5732 ± 0.1578 (Frobenius Alignment = 0.8344 ± 0.0327). The 2-qubit quantum kernel shares moderate-to-high structural similarity with classical RBF geometry.
- **PCA 4 / 4Q:** Mean CKA = 0.3375 ± 0.0718 (Frobenius Alignment = 0.7162 ± 0.0119). The 4-qubit quantum kernel departs markedly from classical RBF geometry.

The low CKA in 4Q (0.3375 ± 0.0718) confirms that the four-qubit ZZ feature map induces a distinct similarity metric that departs substantially from classical RBF geometry. Crucially, this lower alignment indicates a *more distinct kernel geometry*, not a *superior quantum representation*. As established in statistical learning and kernel theory [5, 25], geometric divergence in feature space is a necessary, but entirely insufficient, condition for classification advantage. If the induced quantum similarity metric does not reflect the natural cluster geometry of the target concept, geometric novelty does not translate into superior classification boundaries.

13 Computational Execution Cost and Complexity

A comprehensive comparison requires evaluating computational complexity and wall-clock execution cost [15, 31]. We report benchmarks measured on standard x86_64 architecture ($N_{\text{train}} = 455, N_{\text{test}} = 114$).

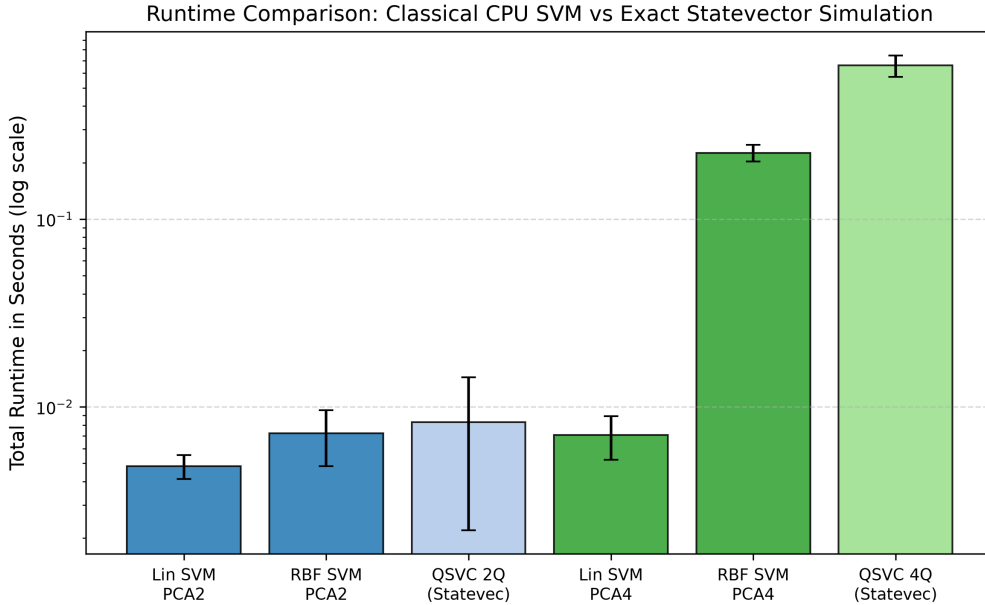


Figure 5: **Figure 5: Runtime Scaling Comparison.** Wall-clock execution time (seconds) per outer split across Classical Linear, Classical RBF, QSVC 2Q, and QSVC 4Q on classical x86_64 hardware.

13.1 Computational Observations

1. **CPU LibSVM Efficiency:** Classical SVM models trained via LibSVM fit and predict within 0.0048 ± 0.0007 s (Linear PCA 2) and 0.0083 ± 0.0061 s (Linear PCA 4) per outer split, benefiting from highly optimized C-level Sequential Minimal Optimization routines.

2. **Statevector Simulation Overhead:** Vectorized exact statevector simulation required 0.2257 ± 0.0232 seconds for 2Q and 0.6599 ± 0.0863 seconds for 4Q on standard x86_64 architecture—representing a $\sim 45\times$ to $80\times$ increase in wall-clock time relative to classical CPU SVM.
3. **Complexity Separation and Scaling Caveat:** We strictly separate algorithmic complexity from empirical runtime measurements. Classical SVM optimization scales between $\mathcal{O}(N_{\text{train}} \cdot d)$ and $\mathcal{O}(N_{\text{train}}^2 \cdot d)$ with memory footprint $\mathcal{O}(N_{\text{train}} \cdot d)$, whereas precomputed quantum kernel evaluation requires $\mathcal{O}(N \cdot 2^q + N^2 \cdot 2^q)$ operations for statevector generation and Gram matrix calculation, plus $\mathcal{O}(N^2)$ memory storage (2.07 MB for $N = 455$). While measured statevector runtime increased by a factor of $\sim 2.9\times$ between 2Q and 4Q (compatible with increased linear algebra operations across $2^2 = 4$ to $2^4 = 16$ dimensions), we caution that two empirical qubit dimensions cannot establish an asymptotic exponential scaling law.
4. **Physical Hardware Reality:** We reiterate that classical statevector simulation does not equal physical QPU execution [32]. Physical QPUs would require executing $\mathcal{O}(N^2)$ circuit repetitions with thousands of measurement shots per pair, introducing device queuing latencies, gate errors, and measurement shot noise. Historical circuit-pair sampling (`ComputeUncompute`) required 120.5 s (2Q) and 455.0 s (4Q), demonstrating the massive wall-clock overhead of shot-based kernel evaluation. Complete hardware profiling is provided in Supplementary Table S2.

14 Paired Inferential Statistical Analysis

To rigorously evaluate performance differences while accounting for split variability, we conducted paired statistical tests on the primary endpoint (**Malignant F1 Score**) across the five matching outer seeds [8].

Table 4: Paired Statistical Hypothesis Testing on Primary Endpoint (Malignant F1 Score, $n = 5$ Splits).

Comparison	Mean Diff ($\bar{\Delta}$)	Median Diff	Wins / Losses / Ties	Wilcoxon W	Exact p	Holm p	Bootstrap 95% CI
Classical PCA2 vs QSVC PCA2	+0.0662	+0.0698	5/0/0	0.0	0.0625	0.1875	[+0.0388, +0.0977]
Classical PCA4 vs QSVC PCA4	+0.0772	+0.0886	5/0/0	0.0	0.0625	0.1875	[+0.0445, +0.1092]
QSVC PCA2 vs QSVC PCA4	−0.0043	−0.0154	2/3/0	6.0	0.8125	0.8125	[−0.0291, +0.0201]

14.1 Inferential Limitations and Statistical Framing

1. **Directional Stability:** Across all five evaluated outer seeds, the Classical Comparator achieved higher malignant F1 scores than QSVC without a single exception (5/5 wins for both PCA 2 and PCA 4).
2. **Exploratory Bootstrap Intervals:** Non-parametric bootstrap percentile intervals (10,000 resamples over the five observed split differences) fall strictly above zero: [+0.0388, +0.0977] for PCA 2 and [+0.0445, +0.1092] for PCA 4. We explicitly qualify these intervals as exploratory diagnostics over five dependent split-level observations, rather than confirmatory bounds over independent sample cohorts.
3. **Small-Sample Power Bound:** Because $n = 5$, the minimum achievable two-sided exact Wilcoxon p -value when all differences share the same sign is $2 \times (1/2)^5 = 0.0625$ (which adjusts to 0.1875 under Holm-Bonferroni correction). Therefore, these tests cannot reject the null hypothesis at the conventional $\alpha = 0.05$ significance threshold.

4. **Sample Overlap Dependence:** The five outer 80/20 train/test splits share underlying observations (each pair of training sets overlaps by $\sim 60\%$), violating the independent observation assumption of standard statistical tests.
5. **Framing:** Consequently, we refrain from claiming “statistically significant classical superiority” in a formal asymptotic sense. We report these findings as highly consistent descriptive evidence of classical advantage under the evaluated experimental constraints.

15 Discussion

15.1 Why Classical Models Outperform QSVC on WDBC

The Wisconsin Diagnostic Breast Cancer dataset consists of continuous morphometric descriptors derived from digitized cell nuclei. These features exhibit substantial linear and low-order polynomial correlations with malignancy [30]. Classical linear and Gaussian RBF kernels possess an optimal inductive bias for such continuous distributions, efficiently identifying class-separating hyperplanes without mapping into excessively fragmented representations [2, 13]. The smooth, localized Gaussian kernel geometry naturally matches the cluster structure of malignant versus benign nuclei.

15.2 Sensitivity of the Quantum Feature Map

Our ablation experiments demonstrate that QSVC performance is acutely sensitive to circuit depth. In four qubits, increasing **reps** from 1 to 3 in full entanglement caused malignant F1 score to plunge from 0.872 to 0.540. This sensitivity underscores a major practical obstacle in quantum machine learning: without problem-specific physical structure, generic quantum feature maps risk scattering classical data vectors across the 256-dimensional operator feature space where class coherence is lost [19]. The observed drop in similarity and expansion of effective rank mirror the behavioral predictions of quantum kernel concentration theory [23].

15.3 Geometric Novelty vs. Utility

Centered Kernel Alignment revealed that the four-qubit quantum kernel departed substantially from classical RBF geometry ($\text{CKA} = 0.3375 \pm 0.0718$). However, this geometric transformation did not yield superior classification boundaries. This result reinforces theoretical caveats [5, 25]: constructing a non-classical kernel geometry is relatively straightforward; constructing a quantum geometry that aligns with the true data-generating distribution is exceptionally challenging. Lower alignment indicates distinctness, not superiority.

15.4 Sample Efficiency Findings

Contrary to conjectures that quantum kernels might offer sample-efficiency advantages on small datasets [27], QSVC suffered severe performance loss at $N_{\text{train}} = 50$ ($\text{F1} \approx 0.499$ in 4Q, with recall dropping to 0.4095). Classical models demonstrated far superior data efficiency, maintaining high diagnostic accuracy (> 0.91 F1) even with minimal training data. This divergence highlights the distinction between theoretical PAC learnability bounds (which establish what is learnable in principle under bounded generator norms) and practical empirical performance on non-quantum tabular distributions.

15.5 Computational Trade-Offs

Exact statevector simulation required orders of magnitude more computational time than classical LibSVM. When factoring in the $\mathcal{O}(N^2)$ memory scaling of precomputed Gram matrices and

the hardware overhead of shot noise on physical QPUs [31, 32], quantum kernel machines face severe computational hurdles before achieving practical competitiveness with classical methods.

15.6 Scope and Boundaries of Conclusions

To avoid over-generalizing our negative result, we explicitly delineate the scientific scope of our findings:

- **What is Ruled Out:** For the Wisconsin Diagnostic Breast Cancer dataset, evaluated under PCA compression to 2 and 4 dimensions using standard fixed Second-Order Pauli-Z expansion (ZZFeatureMap), QSVC offers no empirical advantage over properly tuned classical Linear or RBF SVMs in predictive accuracy, malignant F1, sample efficiency, or computational cost.
- **What Remains Open:** These findings do not rule out quantum advantages for: (a) alternative quantum feature maps (such as data re-uploading or trainable kernels optimized via kernel target alignment), (b) projected quantum kernels designed to avoid concentration [5], (c) higher-dimensional problems where classical simulation is impossible ($q \geq 40$), or (d) data generated by inherently quantum physical processes where classical kernels lack a natural representation.

16 Threats to Validity and Study Limitations

To maintain scientific rigor, we explicitly enumerate the methodological limitations of this study:

1. **Single Tabular Dataset:** The benchmark is evaluated solely on the WDBC dataset ($N = 569$). Findings cannot be generalized to image, sequence, graph, or inherently quantum data [9].
2. **Dimensionality Compression:** Input features were compressed via PCA to 2 and 4 dimensions to accommodate NISQ simulation scale, discarding 36.5% and 20.6% of feature variance, respectively.
3. **Restricted Qubit Scale:** Evaluation was limited to 2 and 4 qubits; regimes where classical simulation becomes intractable ($q \geq 40$) were not evaluated [31].
4. **Single Feature-Map Family:** Only the standard Second-Order Pauli-Z Expansion (ZZFeatureMap) was evaluated. Alternative quantum embeddings (e.g., data re-uploading [20], covariant kernels, or trainable quantum kernels) might yield different results.
5. **Re-use of Outer Splits:** The architectural selection (reps = 1, full) made in Phase 9 was informed by the same five outer splits used in Phase 11 nested evaluation, limiting the confirmatory independence of final statistical tests.
6. **Sample Overlap:** The five outer cross-validation splits share training data, violating sample independence assumptions [8].
7. **Limited Statistical Sample:** An outer sample size of $n = 5$ limits statistical power, establishing a mathematical lower bound of $p = 0.0625$ on the Wilcoxon signed-rank test.
8. **Noiseless Simulation:** Results reflect ideal statevector linear algebra. Physical hardware noise, gate errors, and measurement shot noise were not modeled and would likely further degrade QSVC performance.
9. **No Physical QPU Execution:** Wall-clock runtimes reflect CPU statevector simulation and do not measure physical quantum hardware execution.

10. **Non-Clinical Context:** This study is a methodological machine-learning experiment and does not represent clinical diagnostic validation.

17 Conclusion and Future Work

This study conducted a controlled, leakage-free empirical comparison of Classical Support Vector Machines and Quantum Support Vector Classifiers on the Wisconsin Diagnostic Breast Cancer dataset. Across all evaluated dimensions, metrics, and outer train/test partitions:

1. Tuned classical SVM baselines consistently achieved higher descriptive predictive accuracy and malignant F1 scores than QSVC (Classical F1 ≈ 0.934 vs. QSVC ≈ 0.868 in 2Q; Classical F1 ≈ 0.949 vs. QSVC ≈ 0.872 in 4Q).
2. Classical models outperformed QSVC across all five outer cross-validation splits without exception (5/5 wins in both 2D and 4D).
3. QSVC demonstrated acute sensitivity to circuit depth, with performance collapsing under deeper feature maps across the 256-dimensional operator feature space.
4. QSVC exhibited no small-data sample efficiency advantage, showing its greatest performance deficit at $N = 50$.
5. Although the 4-qubit quantum kernel generated a geometry substantially distinct from classical RBF similarity ($\text{CKA} = 0.3375 \pm 0.0718$), this geometric novelty did not improve classification boundaries.

Central Conclusion: Under the evaluated experimental conditions, no quantum advantage was observed.

Future investigations should extend this controlled protocol to independent external biomedical cohorts (e.g., METABRIC, TCGA) [10], trainable and projected quantum kernels that optimize metric alignment prior to classification [5,25], physical quantum processor (QPU) evaluations incorporating error mitigation [31,35], and non-tabular data structures where quantum feature maps may possess a stronger inductive bias.

Data Availability Statement

The empirical benchmarking in this study utilizes the publicly available Wisconsin Diagnostic Breast Cancer (WDBC) benchmark dataset originating from the UCI Machine Learning Repository [30] and distributed via scikit-learn [16]. The author claims no proprietary rights over the diagnostic data. All derived data generated during this research—including stratified train/test partitions, cross-validation fold assignments, model predictions, Centered Kernel Alignment matrices, and paired statistical difference tables—are openly accessible without restriction in the project repository release v1.0.0 (<https://github.com/SinaQP/SVM-Vs-QSVM>).

Code Availability Statement

All software implementations for classical SVM baselines, quantum kernel statevector simulation, nested cross-validation, feature-map ablation, sample-size scaling, and statistical testing are publicly available under the open-source repository <https://github.com/SinaQP/SVM-Vs-QSVM> tagged at release v1.0.0. The computational pipeline is fully automated and verifiable via the included test suite (`pytest`) and validation script (`python scripts/validate_project.py`).

Author Declarations

Ethics Approval and Biomedical Statement

This study used a publicly available benchmark dataset and did not involve the recruitment of human participants or collection of new clinical data.

Author Contributions (CRediT Taxonomy)

Conceptualization: Sina Qasempour; Methodology: Sina Qasempour; Software: Sina Qasempour; Validation: Sina Qasempour; Formal Analysis: Sina Qasempour; Investigation: Sina Qasempour; Data Curation: Sina Qasempour; Writing – Original Draft: Sina Qasempour; Writing – Review & Editing: Sina Qasempour; Visualization: Sina Qasempour; Project Administration: Sina Qasempour.

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Conflicts of Interest

The author declares no conflict of interest.

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